# Eighth Implementation Iteration: Space Allocation

## with Space Availability Status

Software Design / Iteration Report

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## 1. Iteration Goal

The goal of this eighth implementation iteration is to extend the prototype by adding a simple availability status for each space. In previous iterations, the system evaluated requests according to participant count, required features, and time-based availability. However, it could not represent the case where a space is temporarily unusable due to maintenance.

In this iteration, each space is given an operational status indicating whether it is currently available. A request is approved only if the requested space satisfies the capacity requirement, the required feature requirement, the availability status requirement, and the day/time availability rule.

## 2. Changes from Iteration 7

Compared with the seventh iteration, the following changes were introduced:

- Extended the `Space` model by adding an availability status field
- Added a new decision rule: `StatusRule`
- Updated the request-processing logic so that a request is rejected if the requested space is under maintenance
- Preserved the same request types:
  - `OneTimeRequest`
  - `RecurringRequest`
- Preserved the same supported space types:
  - `Classroom`

  - `Laboratory`
  - `MeetingRoom`

- Preserved the same existing rules:
  - `AvailabilityRule`
  - `CapacityRule`
  - `FeatureRule`

## 3. Scenario

A user submits either a one-time request or a recurring request for a specific space resource. Each request may include a required feature such as a projector, whiteboard, or computers.

In this iteration, the system first checks whether the requested space is currently available for use. If the requested space is under maintenance, the request is rejected immediately. Otherwise, the system continues by checking the required feature, capacity, and day/time availability conditions.

A one-time request is approved if its single occurrence passes all rules. A recurring request is approved only if all occurrences pass all rules. Otherwise, the request is rejected.

## 4. Modules and Their Tasks

### A. Request Management Module

- Receive one-time or recurring space requests from the user
- Create and store `OneTimeRequest` or `RecurringRequest`
- Include request details:
  - requester
  - requested space
  - requested time slot or time slots with day information
  - participant count
  - required feature
- Send the request to the Rule Evaluation Module

### B. Resource Management Module

- Manage all supported space types
- Provide information for `Classroom`, `Laboratory`, and `MeetingRoom`
- Provide identity, type, name, capacity, supported features, and current availability status of the requested space

### C. Rule Evaluation Module

- Apply `CapacityRule`
- Apply `FeatureRule`
- Apply `StatusRule`
- Apply `AvailabilityRule`
- For recurring requests, evaluate all requested occurrences
- Return the final decision: Approved or Rejected

### D. Allocation Management Module

- Create one allocation for a one-time request if all rules pass
- Create multiple allocations for a recurring request if all occurrences pass
- Update request status to Approved or Rejected

## 5. Iteration Comparison

| **Iteration** | **Request Types** | **Resource Types** | **Rules Implemented** |
|---|---|---|---|
| Iteration 1 | OneTimeRequest | Classroom | AvailabilityRule |
| Iteration 2 | OneTimeRequest | Classroom | AvailabilityRule + CapacityRule |
| Iteration 3 | OneTimeRequest | Classroom + Laboratory | AvailabilityRule + CapacityRule |
| Iteration 4 | OneTimeRequest | Classroom + Laboratory + MeetingRoom | AvailabilityRule + CapacityRule |
| Iteration 5 | OneTimeRequest + RecurringRequest | Classroom + Laboratory + MeetingRoom | AvailabilityRule + CapacityRule |
| Iteration 6 | OneTimeRequest + RecurringRequest | Classroom + Laboratory + MeetingRoom | AvailabilityRule + CapacityRule |
| Iteration 7 | OneTimeRequest + RecurringRequest | Classroom + Laboratory + MeetingRoom | AvailabilityRule + CapacityRule + FeatureRule |
| Iteration 8 | OneTimeRequest + RecurringRequest | Classroom + Laboratory + MeetingRoom | AvailabilityRule + CapacityRule + FeatureRule + StatusRule |

## 6. Updated Class Diagram

Figure 1 shows the updated simplified class diagram for the eighth implementation iteration.

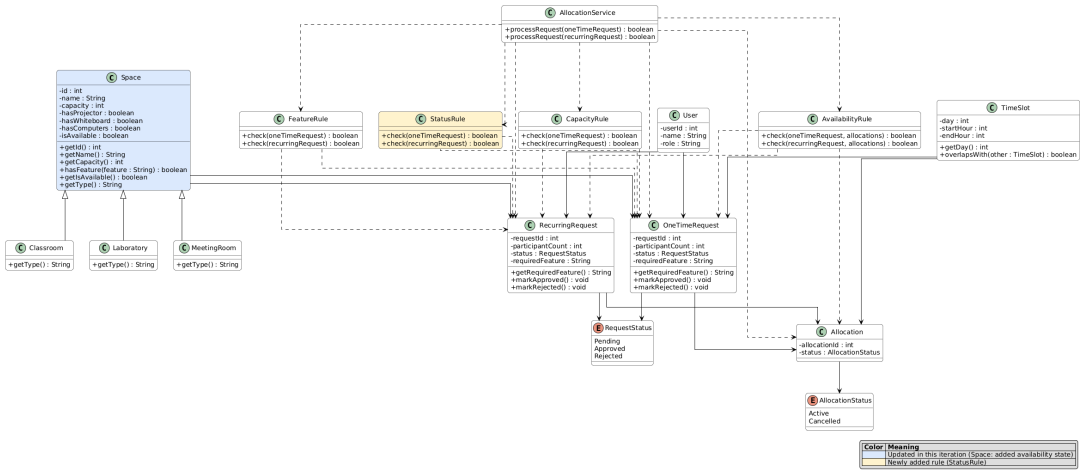


Figure 1: Updated Simplified Class Diagram

# 7. Updated Workflow Diagram

Figure 2 shows the workflow for the eighth iteration of the space allocation process.

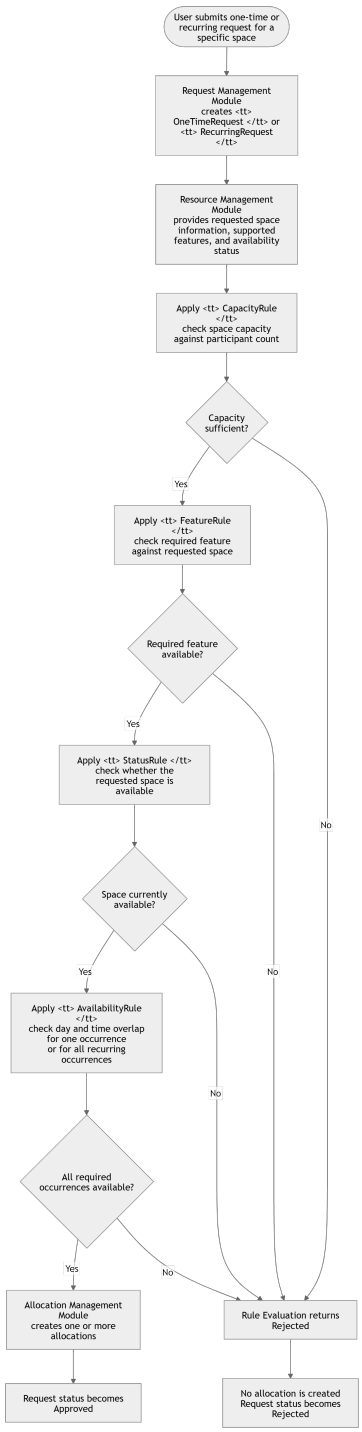


Figure 2: Updated Workflow Diagram

## 8. Prototype Implementation

To validate the design of this iteration, the C++ prototype was extended by adding an availability status field to the `Space` model. A new decision rule, `StatusRule`, was introduced to verify that the requested space is currently available before allocation is considered.

The allocation service continues to support both one-time and recurring requests. It now checks the capacity rule, the feature rule, the status rule, and the availability rule before creating allocations.

If all rules pass, one or more allocations are created depending on the request type. If one or more rules fail, the request is rejected.

## 9. Summary

This eighth implementation iteration extends the prototype by introducing a simple availability status for each space while preserving the existing request types, space types, and allocation architecture. The system can now reject requests for spaces that are temporarily unavailable due to maintenance.

This iteration demonstrates controlled iterative development by adding another small rule-based constraint without increasing workflow complexity. Future iterations can continue this approach by refining space management or adding simple decision explanations.